

OSHAN CHAMARA

☎ +94 76 343 7235 ✉ oshanrajapaksha98@gmail.com [in linkedin.com/in/oshan-chamara](https://www.linkedin.com/in/oshan-chamara)
github.com/oshan1998

Profile

Software Engineer with 2+ years of experience building backend systems, full-stack applications, and AI-powered solutions. Experienced in owning software projects end-to-end, from requirements and architecture through development, testing, deployment, and production support. Strong in TypeScript, Python, Node.js, React, NestJS, cloud infrastructure, and LLM integrations, with a focus on building reliable and maintainable software.

Skills

Professional: Problem Solving, Team Collaboration, Client Communication, Technical Documentation
Engineering: Full-Stack Engineering, RESTful APIs, WebSocket, API Integration
Development: TypeScript, JavaScript, Python, Node.js, NestJS, React, Angular, FastAPI, Flask
Infrastructure: MongoDB, BigQuery, AWS, GCP, Docker, Nginx, PM2, Git, GitHub, Jira, Confluence
AI: OpenAI APIs, Gemini, MCP, AI/LLM Integration

Experience

ZOOMi Technologies Inc.

Full-Stack Software Engineer

Jul 2024 – Present

Boralasgamuwa, Sri Lanka

- Took full ownership of a **live legacy application**, independently maintaining and evolving the system through delivering new features, optimizing data-intensive workflows, resolving production issues, and managing production deployments.
- Owned **end-to-end customer delivery**, from requirements analysis and Jira planning through development, testing, deployment, and production support, consistently meeting agreed timelines.
- Architected and delivered a **NestJS/React platform** for a 3D printing company, streamlining its production pipeline through **technician task assignment, job tracking, and status monitoring** within **three months**.
- Designed and built a **multi-source transaction synchronization service** maintaining transaction history and running balances in **BigQuery** for automated **QuickBooks reconciliation** by AI agents.
- Designed and implemented a **source-agnostic master data synchronization service** with hash-based CDC, keeping BigQuery datasets current for downstream AI agents.
- Designed and built a queue and scheduler-based **control plane** for custom QuickBooks ETL pipelines, completely replacing Fivetran for BigQuery synchronization.

Tech: NestJS, React, Angular, MongoDB, Python, FastAPI, BigQuery, AWS, GCP, OpenAI, Docker

Vega Innovations (Pvt) Ltd

Trainee Electronic Engineer

Jan 2023 – Jun 2023

Colombo, Sri Lanka

- Designed an EEPROM library with wear leveling, used in production firmware of the **Vega ETX** electric three-wheeler.
- Extended and maintained the display firmware for Vega ETX and the Vega conversion kit.
- Modified the SPC5 custom bootloader of the Vehicle Control Unit, designing an algorithm to handle errors during OTA firmware updates.
- Designed the PCB and firmware for a rotary gear knob with an LCD display for an upcoming electric mini car; ported an Arduino IMU sensor library to STM32 and SPC5.

Tech: C/C++, STM32, SPC5, CAN, SPI, QSPI, UART, Eagle PCB

Projects

AXN - Agent Workflow Graph Platform | *NestJS, MongoDB, MCP, Vertex AI* Jun 2026 – Aug 2026

- Designed and developed a **graph-based agent workflow runtime** from scratch, enabling complex multi-agent systems to be composed and executed as reusable workflow graphs without relying on frameworks such as LangGraph or LangChain.
- Built a configurable execution engine supporting **node dependencies, context propagation, parallel execution, retries, failure handling, and workflow state persistence**.
- Implemented and tested agent networks for **ERPNext business process automation, AI content generation, and multi-step autonomous workflows**.

AgentX - Hierarchical AI Agent Runtime | *Node.js, TypeScript, LLMs, Vertex AI* May 2026

- Designed and built a **modular hierarchical AI agent runtime** from scratch, where a principal agent plans tasks and delegates work to isolated sub-agents.
- Implemented a **parallel DAG-based task orchestrator** with sandboxed sessions, scoped tool permissions, and configurable iteration budgets.
- Built a **pluggable skill/tool architecture** with multi-provider LLM support (Gemini, OpenAI, Ollama), session and long-term memory, scheduling, and real-time WebSocket tracing.

Out of Distribution Generalization [FYP] | *PyTorch, Knowledge Distillation, Cross Attention* Apr 2024

- Proposed an **invariant feature learning** method for OOD classification using image captions and **cross-attention-based latent feature alignment**.
- Used **knowledge distillation** to transfer invariant representations to an image classification encoder.

Education

University of Moratuwa

B.Sc. Engineering (Hons.), Electronic and Telecommunication Engineering

- **Second Class Upper Division**

Jan 2020 – June 2024

Moratuwa, Sri Lanka

Certificates

- Machine Learning Specialization (3 courses) – DeepLearning.AI
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning – Coursera

Achievements

- 5th Place :- Hackerrank Coding Competition for Data Structures and Algorithm Module
- Semi Finalist :- Future Innovators Challenge '22
- Top 10 :- Spark Challenge 2021/22

Interests and Hobbies

- Interests :- Agentic AI, Distributed Systems, System Design, Electronics, Robotics
- Hobbies :- Listen to the music, Cricket, Carrom, Swimming

References

Mr. Isuru Avishka Deshan

Technical Team Lead

ZOOMi Technologies Inc, Sri Lanka

+94 77 292 0437 | isuru@zoomi.ca

Mr. Nisal Bulathsinghala

Associate Technical Team Lead

Vega Innovations (Pvt) Ltd, Sri Lanka

+94 77 766 7795 | nisal@vega.lk